

Augmented Reality-based Indoor Positioning for Smart Home Automations

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ABSTRACT

Ambient Assisted Living (AAL) has been discussed for some time; however, many systems are not considered as interoperable or user-friendly. This fact is an even more important issue as every interaction is more costly in terms of time and effort for people with disabilities or senior citizens. Therefore, this paper examines the potential of automations that can substitute typical daily interactions in AAL or Smart Home settings in general based on the users' location. Particularly, we suggest the novel approach of using the indoor positioning capabilities of Augmented Reality (AR) head-mounted displays (HMD) to detect, track, and identify residents for the purpose of automatically controlling various Internet of Things (IoT) devices in Smart Homes. An implementation of this feature on an off-the-shelf AR HMD without additional external trackers is demonstrated and the results of an initial feasibility study are presented.

CCS CONCEPTS

• **Human-centered computing** → Accessibility; Accessibility technologies; Human computer interaction (HCI); Human computer interaction (HCI); Interaction paradigms; Mixed / augmented reality.

KEYWORDS

Indoor Positioning, Augmented Reality

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1 INTRODUCTION

Smart Home systems are gaining popularity and it becomes easier to integrate solutions from various vendors thanks to new open standards such as Matter [34, 38]. However, Smart Home systems equipped with smart lights, smart thermostats, or smart speakers require meaningful data input to enable smart automations.

Without linking actors to sensor data, these systems remain anything but “smart,” and users are forced to detour through opening vendor-specific smartphone apps to perform even simple tasks. One promising source to trigger automations is data about a user's location. The room temperature, for instance, can be automatically lowered to save energy when all residents leave the house. A system that is aware of its users' indoor positions can provide even more automated features regarding, e.g., user comfort, safety, or security. Thereby, these automations may not only increase convenience for healthy residents at home, but also allow senior citizens and people with disabilities to regain autonomy and improve their quality of life.

Current Smart Home systems usually solve the task of detecting users with the help of passive infrared (PIR) motion sensors. Although being affordable, PIR sensors suffer from several drawbacks [20]. Besides PIR sensors, more advanced so-called indoor positioning systems (IPS) are available, but they are mostly intended for professional applications. Still, most IPS need – depending on room size and required resolution – one or more sensors and/or receivers per room, which often leads to expensive solutions with the need for professional installation and setup.

In this paper, we shed light on a future scenario by looking at the positioning information obtained through an Augmented Reality (AR) head-mounted displays (HMDs) with tracking capabilities across six degrees of freedom (6DoF). While such devices are already available and can be used for research today, they are still bulky, expensive, and not suitable for everyday use. Several companies such as Apple [17], Meta [13], and Bosch [6] follow and invest into the vision of “Ubiquitous Augmented Reality” [32]. Therefore, we make the assumption that unobtrusive AR HMDs will be available in the near future for this scenario and that users will wear these HMDs all-day as a potential replacement for smartphones and smartwatches. 6DoF AR HMDs are equipped with the necessary sensors to accurately position themselves within a house or an apartment to reliably anchor virtual objects in a room. Prior research has used this ability for instance for indoor navigation in public buildings [24].

We suggest applying AR self-positioning to the Smart Home domain and its automations. Thus, reliable user tracking in a Smart Home environment without the need to install and setup dedicated external devices is available as soon as user-friendly, i.e., lightweight, comfortable, and affordable AR HMDs become feasible.

2 RELATED WORK

Smart Home systems have extensively been researched in the past years [25]. In general, there are a wide variety of application areas

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within the Smart Home context, such as safety, sustainability, security, comfort, or entertainment [25]. Especially their application for senior citizens or people with disabilities as Ambient Assisted Living (AAL) systems has been of high interest due to the aging population and rising costs of healthcare [35]. AAL provides assistive tools such as health and activity monitoring or fall and wandering prevention [35].

2.1 Indoor Positioning Systems in Smart Homes

Many Smart Home applications require or can be substantially improved by referring to the residents' position. For example, Smart Home systems detecting room occupancy can control lighting or heating, ventilation, and air conditioning (HVAC) more energy-efficiently [1, 46]. Additionally, alarm systems can differentiate between residents and intruders [19]. In the context of AAL, activities of daily living can be monitored [22, 39, 46], abnormalities detected in combination with vital sensors [30, 37], and care givers can be relieved for example by notifying them only for tasks where help is required [2]. In general, this means that more affected people can remain living at home, which can increase the quality of life [36].

Depending on the technology, IPS can detect, track, or even identify residents at home [8]. *Active* IPS use tags or devices that are carried by the user while *passive*, device-free systems do not require the user to carry anything [1]. Solutions are based on different technologies such as infrared radiation, radio frequency, visible light, physical excitation, computer vision, or light detection and ranging (LiDAR) [1, 45]. Existing infrastructure based on, e.g., Wi-Fi or Bluetooth can be used as IPS to reduce costs, however, this may have adverse effects on the primary purpose of the technology, i.e., data transmission [45].

While passive and unobtrusive positioning is more user-friendly, these systems come with many challenges especially when tracking or identifying multiple subjects [1, 8]. Complex systems typically require professionals for setup and maintenance [2]. Hence, smart floors with a dense network of sensors underneath the floor, cameras that require proper lighting and can suffer from occlusion and blind spots, or systems sensible to environmental changes [43] can become problematic [1]. From a practical standpoint, [45] call for system requirements such as minimal calibration, resilience against environmental changes (e.g., moving furniture), real-time positioning, and low computational complexity and energy-consumption with low costs at the same time. Especially for senior citizens, usability, acceptance, and safety are named as the most important factors [30]. However, general acceptance and interoperability of AAL systems still appears to be an issue [39].

2.2 Augmented Reality-based Indoor Positioning Systems

AR devices need to acquire their pose in 3D space to display and anchor virtual objects aligned with the physical environment [5]. This self-positioning is achieved by applying simultaneous localization and mapping (SLAM) algorithms often based on different sensors to create a 3D representation of the environment and derive the users' location within this environment at the same time [10, 29]. As satellite-based positioning systems (e.g., GPS) are not

available indoors, numerous research projects have considered the self-positioning feature of AR combined with the ability to display spatial information to create AR indoor navigation systems to find a destination in an unknown environment such as a hospital [9, 16], a university [18, 40], provide help for people with low vision [7], or an escape route in an emergency situation [44]. Additionally, positional information can be used to display context-aware information in museums [23] or to support workers in factories [14] or facility management [4].

AR as an interface for Internet of Things (IoT) devices in Smart Homes has been studied by several authors lately [11, 15, 27, 33, 41, 42]. Most of these studies focus on novel 3D-based interaction modes to control these devices, display information at their physical location, or create automations between them. However, prior research has not yet considered the potential of using the users' position for Smart Home automations, which can substitute certain user interactions altogether and actually integrate the user in their Smart Home environment.

3 METHODS AND APPROACH

To explore the technical and practical feasibility of AR HMDs for indoor location-based Smart Home automations, we derived requirements from prior research and own considerations to build an early prototype.

3.1 Design Considerations

The user tracking requirements for Smart Home automations vary depending on the use case. A higher resolution enables Smart Home systems to recognize user intents more accurately and ensure that automations are not accidentally triggered. Moreover, the AR HMDs should be able to also track themselves in dark environments for example to turn on lights when entering a room. Additionally, automations need to be aware of every resident to avoid turning off lights in a room that is still occupied when one person leaves. In general, the assumption of everyone wearing an AR HMD at all times for accurate tracking depends on the attributes of the headset, such as weight, comfort, or runtime. Still, a fallback mechanism is required when guests are visiting or small children without AR HMDs are living there. Thus, this solution should be implemented as part of a multi-modal approach and regular switches are still useful.

3.2 Prototype Description and Introduction of Use Cases

For our prototype implementation, we chose to use a 6DoF AR HMD with a high resolution and pose the assumption that residents wear the HMD most of the time to enable and test various novel use cases. Therefore, we selected the Microsoft HoloLens 2 as AR HMD for this prototype as it represents the state of the art in AR technology and supports these requirements. In the first step, we use the HoloLens 2 to create a 3D scan of the test apartment. The scan is imported and authored in Unity 2020.3.34f. To align the virtual scan and the physical environment, we use multiple spatial anchors from Microsoft's World Locking Tools that enable cross-device and cross-session consistency. Each room and more specific areas of interest are marked with boxes (colliders). Entering

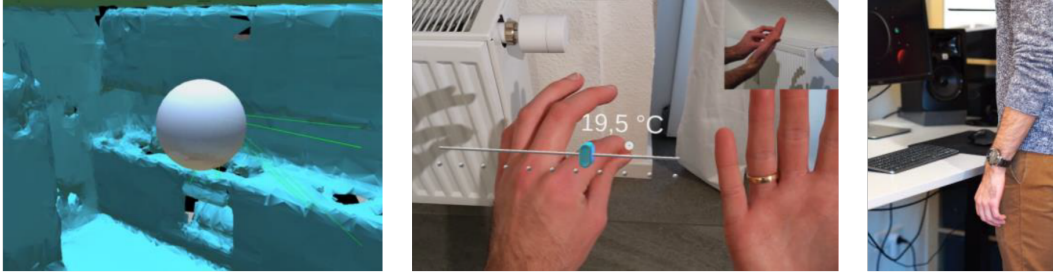


Figure 1: Three potential use cases for location-based Smart Home automations. Left to right: Tracking resident position within the kitchen to turn on countertop lighting when approached (left); automatically selecting the IoT devices in the current room for user interactions (middle); moving height-adjustable desk to user's standing or sitting position (right).

or exiting a collider publishes an event to the Smart Home system to notify it about the user's position. The Smart Home system for this prototype is a Home Assistant¹ v2023.1.2 instance that controls off-the-shelf IKEA smart lights and Tado thermostats. While the AR HMD only sends location events similar to a motion sensor (i.e., inputs), Home Assistant handles the resulting actions in automation scripts (i.e., processing). In the following, we introduce three different potential future use cases that rely on room-level and centimeter-level detection, tracking, and identification.

Overall, the user can interact with the system as follows: For the first use case, users call a slider to control the room temperature by looking at their open hand as depicted in Figure 1. The selected temperature setting is automatically matched to the room in which the user is currently located. Thus, users do not have to browse a list of all thermostats and rooms to control the temperature. Additionally, when entering or leaving a room, the system turns the lights on and off. We use a second HoloLens 2 to simulate two residents and their interaction to, e.g., keep the lights on when one person remains in the room. In our prototype, both AR HMDs only communicate through Home Assistant and are not aware of each other. The second use case uses the high spatial resolution to control the countertop lighting in a kitchen. When a resident approaches the countertop, the lights turn on. Moving outside the collider area turns the lights off again. The third use case relies on height and the additional rotation information provided by the AR HMD to control a height-adjustable desk. By standing in front of the desk, it can move up to an ergonomic standing position. In turn, if the user sits down in front of the desk, it moves down to a seated position. To avoid false positives, the mechanism is only activated if the user is oriented towards the desk.²

3.3 Latency comparison with regular PIR motion sensors

The positioning performance potential of camera-based SLAM systems has already been demonstrated [31]. As PIR sensors are already often used in Smart Homes but can only detect and not track or identify residents³, we wanted to compare the latency of a PIR sensor with our prototype for the use case of turning on lights in a

realistic setting. To avoid residents entering a dark room and having to wait until the lights turn on, this operation should take below one second. For the study, a test subject enters a room at around 1 m/s as measured by markers on the floor and a metronome.

We tried an off-the-shelf IKEA Trådfri motion sensor, however, this device has a “recharge time” of around three minutes during which it is blind and does not report motion to save energy. Therefore, we built a non-battery-powered solution based on a standard HC-SR501 motion sensor⁴ connected to an ESP32 microcontroller that runs ESPHome v2022.10.0 and publishes sensor data to the Home Assistant instance via Wi-Fi. Home Assistant is among the largest open-source projects on GitHub in general and in the smart home system domain [12]. The HC-SR501 and the ESP32 were chosen as they are widely available development hardware that support ESPHome and, thus, can be easily integrated into Home Assistant. We measure the end-to-end latency between entering the room and the timestamp of the reported event in Home Assistant when it becomes available for further automation tasks and count false negatives if one system misses an entry event. The motion sensor is placed just behind a corner close to the path of the test subject. For the AR HMD, we defined the boundary of the room at the intersection of the motion sensor's field of view and the middle of the path as depicted in Figure 2.

Since the timestamps of the motion sensor and the AR HMD only provide a relative latency between both systems, we filmed the entering event with a slow-motion camera at 480 fps to obtain more precise data. A LED light was added to the microcontroller to indicate that the motion sensor registered a movement. Thus, we measured the latency as number of frames between the point of entering and the LED turning on in the video. The first part of the study was conducted in a well-lit room from daylight and the artificial room lighting. While the PIR sensor filters visible light, the HoloLens 2 requires a well-lit room [28]. Hence, we controlled for the light in a second part measuring only the latency difference between PIR sensor and AR HMD to test if dark environments hinder the practical application of AR as IPS. The room was solely artificially lit with dimmable lights.

In general, we expect the PIR sensor to react quicker than the AR HMD due to the lower computational complexity of the mechanism. For the second part of the study, we expect the detection latency to

¹<https://www.home-assistant.io/>

²A demonstration of the prototype can be available here: <https://youtu.be/z0OErqb09J8>

³Tracking is possible with a dense array of PIR sensors, e.g., [21].

⁴Datasheet: <https://www.epitran.it/ebayDrive/datasheet/44.pdf>

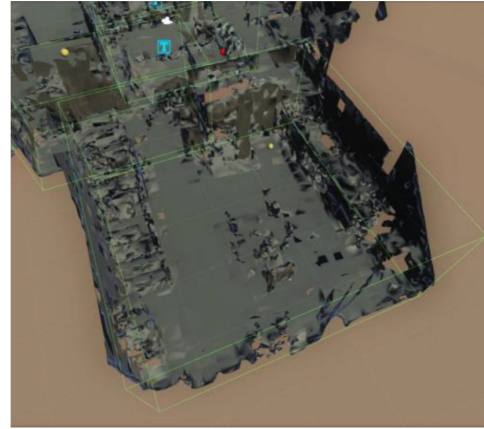
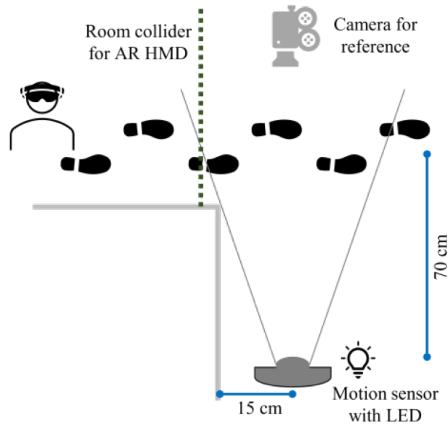


Figure 2: Study setup (left) and the 3D scan of the test apartment with box colliders in green (right)

increase and the HoloLens 2 to fail at tracking itself below a certain lighting threshold.

4 RESULTS

For the first study part and for each lighting condition in the second part, the test subject entered the room ten times. The variance of both systems would require more measurements for an adequate statistical evaluation which was beyond the scope of an initial proof-of-concept study aiming at testing the general feasibility of the approach. Thus, the results (see Figure 3) should be viewed as exploratory and preliminary. In the first part of the study with the slow-motion camera reference, the PIR sensor ($M = 729.09$ msec, $SD = 183.90$ msec) was significantly slower than the AR HMD ($M = 424.26$ msec, $SD = 154.84$ msec; $t(10) = 9.09$, $p < .001$) with the specific positioning depicted in Figure 2. In the second part with artificial lighting only, the latency differences between both systems are distinctly smaller ($\Delta M = 22.71$ msec, $SD = 74.02$ msec) and could also be attributed to variances in data transmission to or data processing in Home Assistant. We used a BH1750 light sensor⁵ connected to a second microcontroller to measure the illuminance. During the first study part, the sensor measured 78 lx. In the second part, we stepwise dimmed the lights from 100% power (measured at 46 lx) to 0% (0 lx) in steps of 20%. At 60% (17 lx) and 20% (4 lx), the HoloLens 2 missed one out of ten entry events. Even at 5% (1.5 lx), the HoloLens 2 could detect entry events in all ten runs and the latency is still comparable to the PIR sensor ($\Delta M = 14.61$ msec, $SD = 35.00$ msec). However, the primary use case of displaying holograms does not reliably work anymore below 40% (8 lx) resulting in a jumping and unstable depiction of the holograms. Finally, when entering a dark room (0 lx) from a dimly lit room, three out of five entry events were missed. Hence, the HoloLens 2 becomes unreliable whereas the PIR sensor detects every entry motion.

5 DISCUSSION AND FUTURE WORK

Benefits. The localization based on AR HMDs brings several advantages for Smart Home users. Compared to PIR sensors and many

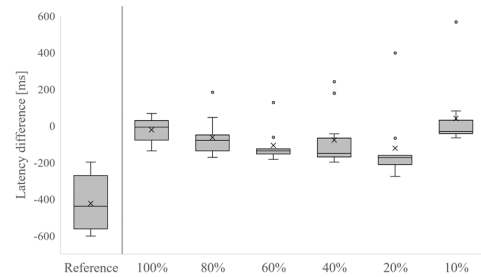


Figure 3: Results of the conceptual study.

IPS alternatives, no external hardware is required to determine the user's exact position. Thus, the solution scales only with the number of users and is independent from living area and number of rooms. As the Smart Home system can differentiate between users based on authentication on the AR HMD or alternatively their height, the system can personalize rooms and user interfaces [26]. Moreover, PIR sensors struggle with detecting non-moving human subjects (e.g., on a couch or in the bathroom) which complicates room occupancy estimation [3]. As IPS know where each resident is, determining the occupancy of each room is trivial and the reliability of related automations is higher.

Use Cases. The positioning resolution of AR HMDs enables a variety of automation use cases in Smart Homes (as listed in Table 1). Accurate room occupancy data does not only allow for exact light controlling but also occupancy prediction to pre-heat or cool rooms to save energy without compromising on user comfort. Music can follow the user around the home by only playing on speakers near them. The aforementioned room personalization can encompass light color and temperature, ambient sound, room temperature, or the art selection of digital picture frames adapted to the user's preferences or the compromise of several simultaneous residents. User locations could also be used for security aspects. Doors could automatically unlock when authenticated users are nearby. Additionally, alarm systems in unoccupied rooms of larger homes could automatically be armed [19].

⁵Datasheet: <https://www.mouser.com/datasheet/2/348/bh1750fvi-e-186247.pdf>

Table 1: Examples of possible location-based automations depending on required resolution and localization type. (1 inch = 2.54 cm)

Required resolution	Localization type	Example
Room-level	Detection	Turn on lights HVAC control
Room-level	Detection and identification	Personalization (lights, media, etc.) Activate alarm system in unoccupied rooms
Centimeter-level	Detection	Open doors Remove potential obstacles such as a vacuum robot (AAL)
Centimeter-level	Tracking	Control specific lights in a room Predict movements to trigger automations before an actual event
Centimeter-level	Detection and identification	Lock and unlock front door
Centimeter-level	Tracking and identification	Set adjustable objects to user preference (e.g., adjustable desk) Recognize Activities of Daily Living (ADL)

AAL literature suggests more use cases based on accurate indoor positioning [39]. Automations that adjust the route of autonomous vacuum cleaners to avoid blocking residents with motor impairments or automatically lock and unlock the bathroom could improve life quality and avoid typical day-to-day inconveniences. In cases of emergency, an AR HMD could display a safe exit route, e.g., based on smoke sensors. Moreover, emergency personnel could access the location of residents if such systems are connected. This support in emergency situations would be even more critical for people with motor impairments.

AR HMDs in the Smart Home open a potential for many more applications. With the principles of universal design, simple improvements cannot only help improve the lives of people with disabilities but also improve comfort and safety for everyone at the same time.

Limitations. The suggested use cases and potentials rely on the availability of a user-friendly AR HMD, which is not yet available today. It might appear far-fetched, yet light-weight, unobtrusive AR devices will likely become available in the near future. Thus, using sensor data from already worn AR HMDs does not pose additional effort or cost and is, therefore, less expensive than using a sensor network, even if the individual sensors are inexpensive.

The HoloLens 2 used for our prototype does not have sensors that sustain its features in dark environments. Thus, when walking into a dark room, the device often loses track of its position and could incorrectly trigger automations for other rooms. Additionally, the rather slow update rate of several tracking sensors and their processing struggles with fast movements of the user. Our exploratory study demonstrated that the camera-based self-tracking performs consistently fast when compared to the popular PIR sensor. Particularly in low-lighting conditions, the performance turned out to be better than expected. The results are also limited to the experimental setting and would be different if the collider would be larger or the PIR sensor would be moved. Nevertheless, additional tracking sensors that do not rely on room illumination should be integrated for AR HMDs to be applicable to realistic home environments. Still, the HoloLens 2 is capable of demonstrating future opportunities and use cases of user-friendly AR HMDs.

Today, Indoor Position Systems are frequently criticized for potential privacy issues [1]. Indeed, privacy can be also a challenge for our concept, especially with high-resolution systems that rely on cameras. However, AR-based systems could request user consensus before positional data leaves the device at all. Settings could be applied depending on location (e.g., no tracking within the bathroom). This aspect is also applicable for public environments where a user might, e.g., allow a shopping mall to access the current location to provide indoor navigation, thereby setting permissions for different places. Therefore, developers of AR HMDs should consider data privacy as a fundamental principle always having the users' interests in mind.

Future Work. In a broader context, we want to not only study holistic AR interfaces for Smart Home and AAL systems that include indoor positioning for automations, but also display user interfaces for appliances and IoT devices that only have limited screen space (washing machines, dish washers, etc.) and allow the user to naturally interact with these devices via gestures, eye movement, or speech. Multi-modal and natural interactions, adaptive and personalized user interfaces, and the interoperability of these systems open exciting future research areas.

6 CONCLUSION

Assuming its availability, the application of AR HMDs for indoor positioning in Smart Homes appears both simpler and superior to current solutions and potential alternatives. Reliable positioning of residents allows for several use cases that may improve life quality and autonomy in AAL contexts, and user comfort and security in Smart Home settings in general. The prototype presented in this paper demonstrates the feasibility and performance of the novel idea of an AR HMD for indoor positioning in the context of Smart Home automations on an off-the-shelf device. Still, a lot of effort is required to miniaturize the technology and make this idea a reality.

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